#ManyDaughters

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Pre-analysis plan

1 Introduction

This pre-analysis plan outlines the strategy that our research team will follow to investigate the impact of having daughters on gender attitudes, political preferences, and gendered behaviors within the household in Germany. Specifically, we address how we plan to test four pre-specified hypotheses:

H1. whether having daughters leads to an increase in support for gender equality;

H2. whether having daughters induces more left-leaning political preferences;

H3. whether having daughters is associated with greater gender equality within the household; and

H4. whether having daughters leads to a larger increase in the support of gender equality for men than for women.

For each hypothesis, we specify the variables of interest, the sample restrictions, the exact regression model and the interpretation of the results. The sample restrictions are common across all four hypotheses. The independent variables, controls, empirical strategy and the interpretation of the results are the same for the first three hypotheses, and differ slightly for H4. Finally, the dependent variable is common for H1 and H4, while H2 and H3 require constructing separate outcomes.

The document is organized as follows. Section 2 provides a brief summary of the data source. Section 3 describes the variables of interest and the necessary recoding of the raw data. Section 4 summarizes the inclusions and exclusion restrictions applied to construct the working sample. Section 5 outlines the empirical strategy. Section 6 describes the interpretation of the results. Section 7 lists the planned result output in terms of tables and figures.

2 Data source and sample

We draw on data from the German Socio-Economic Panel (SOEP), an annually collected, nationally representative panel dataset started in 1984 by the Deutsches Institut für Wirtschaftsforschung (DIW Berlin) and covering roughly 15,000 private households each year. Following the German reunification in 1990,

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the survey expanded to include an additional 2,170 East German households. The representativeness of the data is maintained through periodic refreshment cohorts (Wagner et al., 2007).

The SOEP records detailed information on household composition, education, employment, occupation, earnings, health, life satisfaction, political preferences, and gender norms. To support diverse research, the SOEP has incorporated targeted refreshment samples (e.g., immigrants and high-income households) and specialized modules, and it has been harmonized with international longitudinal surveys, such as the Panel Study of Income Dynamics from the United States, for cross-national comparisons. Finally, the longitudinal design of the data enables panel analyses at the individual and household levels (Goebel et al., 2019).

Sample. We have been provided a 5% random subsample of the SOEP. The observation window is restricted to years 2000–2020, and the provided data files are a subset of the SOEP-Core. The methodological approach described in this pre-analysis plan is designed to ensure scalability to the full dataset.

3 Variables

3.1 Dependent variable

This project will study three different dependent variables. Testing H1 and H4 requires a measure of gender attitudes, for H2 we need a measure of political preferences, and assessing H3 entails using a measure of gender equality within the household. Table 1 contains a summary of the outcome variables used in this analysis.

Gender attitudes. H1 and H4 involve assessing differences in support for gender equality, measured by attitudes. The SOEP includes a set of survey questions related to gender role attitudes. Waves 2012 and 2018 contain three questions related to opinions about women and men's participation in housework, childcare and labor market work:[1]

1. "Children below the age of 3 suffer if their mother works"
2. "Best if man and woman work the same amount so they can share the responsibility"
3. "Children below the age of 6 suffer if their mother works"

These items are similar to a subset of questions within the standard gender role attitude questionnaire developed by the International Social Survey Programme (ISSP), which is widespread in the economics literature on gender norms (e.g., Bertrand et al., 2021; Farré et al., 2022). Specifically, these items are "A pre-school child is likely to suffer if his or her mother works" and "Both the husband and wife should contribute to the household income". We believe that the similarity between the ISSP and the SOEP items makes them valid to capture gender role attitudes.

[1] The specific variable names are `plh0302`, `ply0308` and `plh0298`, with suffix `v1` for the 2012 survey year and suffix `v2` for the 2018 wave.

We use item 2 as the basis for measuring gender attitudes to test H1 and H4, as it most directly addresses gender equality.[2] The two questions attached to this item are responded on Likert scales based on agreement with the statement, ranging from "Strongly agree" to "Strongly disagree". The responses in 2012 are recorded in a four-point scale, while the 2018 questions are responded on a seven-point scale. For comparability across waves, first we transform all responses such that one aligns most with traditional gender norms (disagreement with the statement), and higher values indicate closer alignment with progressive gender norms (agreement with the statement).

As our final outcome, we construct an indicator variable that takes value one if the respondent expresses agreement with a more egalitarian distribution of work between men and women, and zero otherwise. We combine the responses in the two available survey years to construct a single outcome. As a result, higher values of the outcome indicate higher support for gender equality.

Political preferences. The provided SOEP files contain two variables related to political preferences:

1. The first one relates to individual *ideological self–placement*. Namely, respondents are asked: "In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?" (variable `plh0004`). The responses are recorded on an eleven-point Likert scale, ranging from "Far left" to "Far right". This question is included in waves 2005, 2009, 2014 and 2019.

2. The second variable corresponds to *political party affiliation*. In every wave, the SOEP includes a survey question related to the "Political Party Supported" (`plh0012_h`), which includes a list of all major German parties as response options.

In an ideal scenario, we would use both variables to address H2, as one captures self-perceived political preferences while the other reflects enacted political attitudes. However, since we must restrict ourselves to a single outcome, we choose the former. The latter may be less informative in some cities or towns, where voting options may not clearly indicate position on the political spectrum.

Since the outcome of interest is alignment to more left-leaning political preferences, we transform the responses to variable 1 such that zero indicates alignment with far right political views and higher values denote a more left-leaning stance. We then standardize the resulting variable within each wave to account for potential distribution shifts across waves. This yields our outcome of political preferences.[3]

Division of household tasks. To test H3, we use measures of self-reported time spent on housework. Every available wave, the SOEP includes three questions about the average hours spent on housework: one for weekdays, one for Saturdays and one for Sundays.[4] Note that equivalent questions are included

[2] We attempted to construct a more comprehensive measure of gender attitudes by performing a factor analysis on all three items. However, the sample size in our 5% data was too small to support the empirical analysis we propose, resulting in code errors and non-reproducible results. With access to the full dataset, we would favor using the more comprehensive measure.

[3] The effective recoding involves: (1) Subtracting from ten the responses to the raw variable; (2) Subtracting from the resulting variable the wave-specific mean, and dividing it by the wave-specific standard deviation. This results in a variable that, in each wave, is centered around zero and has standard deviation of one.

[4] The exact phrasing of the questions is: "What does your everyday life look like in the current situation? How many hours per normal work day/Saturday/Sunday do you spend on the following activities? Housework (washing, cooking, cleaning)." The responses are recorded in variables `pli0012_h`, `pli0016_h` and `pli0043_h`, for Sunday, Saturday and weekdays, respectively.

for labor market work ("Employment or part-time employment, including journey time"). However, since we must restrict ourselves to a single outcome and this hypothesis focuses on the division of household tasks, we choose to retain only the housework variables.

For each respondent, we compute the total weekly hours in housework by adding up the reported average hours on Saturday, Sunday and weekdays five times. To assess gender equality in time allocation, we construct a variable capturing the share of housework done by women. We do so by restricting the sample to respondents with a valid partner identifier (`parid`) and matching individuals' reported hours with their partners' at each wave. We then compute the *wife share* of hours as follows:

$$\text{wife share of housework} = \frac{\text{woman's housework hours}}{\text{man's housework hours} + \text{woman's housework hours}}$$

This share is continuously distributed between zero and one. Shares equal to 0.5 indicate perfect equality, where wife and husband contribute equally, while larger shares indicate female specialization and smaller shares male specialization. Note that this outcome is defined at the household level, so both partners share the same value.

Since to test H3 we want to measure deviations from an egalitarian distribution of tasks, we construct our final outcome to be a variable capturing the absolute deviation of the wife shares from 0.5:

$$\text{AbsDev} = |\text{wife share of housework} - 0.5|$$

This variable ranges from $[0, 0.5]$, with lower values indicating greater gender equality in housework.

Table 1: List of dependent variables used in the analysis.

	Name	Variable	Description
H1-H4	`id_gnorm_eq`	Best if man/woman work the same to share responsibility	Indicator: agreement
H2	`zLeftRight`	Ideological self-placement	Categorical: Right-Left scale of party of affiliation (right-left)
H3	`AbsDev_Home`	Deviation from egalitarian housework	Continuous [0, 0.5]

Notes: This table summarizes all outcome variables constructed to test the four hypotheses in this project. Column 1 indicates the relevant hypothesis. Column 2 displays the name of the constructed variable, as presented in the result tables. Column 3 explains the information contained in each variable. Column 4 provides a description of the values the variable can take.

3.2 Explanatory variables

All hypotheses examine how having a daughter, rather than a son, affects various outcomes. As we explain below, we focus on individuals observed becoming first-time parents, comparing those with a firstborn girl to those with a firstborn boy. Hence, the first regressor of interest is an indicator equal to one if the respondent has a firstborn girl, and equal to zero whenever the child is a boy. Note that this variable is individual-specific, meaning that it takes the value of one for individuals with daughters, irrespective of whether they have had a child already. We refer to this variable as $\textit{Daughter}_i$. The second regressor of interest is a binary variable for parental status, coded as one for respondents who already have children and zero for individuals who are not yet parents in the current period. We refer to this variable as $\textit{Post}_{i,t}$.

H4 additionally studies whether having daughters has a different effect on women in comparison to men. For this analysis, we add a gender dummy to the set of main explanatory variables, which will take value one if the respondent is a man and zero if she is a woman. We refer to this variable as Man_i.

3.3 Control variables

All specifications control for a number of individual and household level demographic, economic and family characteristics. Table 2 includes a full list of the control variables we consider, as well as the format in which they will be included in the specification and the necessary recoding of these variables. On top of these controls, H1, H2 and H3 also include the gender dummy described above.

Table 2: List of control variables.

Variable	Var. name	Recoding	Format	Values
Sex	`sex`	Binary for being a man	Binary	Man, Woman
Number of children	`sumkids`	Pool 5 children or more	Categorical	{1, ..., 5+} children
Marital status	`pgfamstd`	Binary for being in a couple	Binary	Married, single
Area of residence	`sampreg`	–	Binary	East, West
Age	`gebjahr, syear`	Subtract birth from survey year	Continuous	
Age squared	–	Construct from age	Continuous	
Migration background	`migback`	–	Categorical	None, direct, indir.
Education years	`pgbilzeit`	–	Continuous	
Household income	`hghinc`	–	Continuous	
Employment status	`pgemplst`	Construct indicator variables	Binary	Employed, not emp.
Work experience	`pgexpft`	–	Continuous	
Experience squared	–	Construct from experience	Continuous	

Notes: This table lists the set of observable characteristics that we plan to control for in all specifications. Column 1 outlines the characteristic to be controlled for. Column 2 specifies which SOEP variables are used to construct each variable. Column 3 explains any necessary recoding. Column 4 lists the format of each variable. Column 5 indicates the values considered in the case of binary or categorical variables. Note that categorical variables will be included in the different specifications non-parametrically.

4 Inclusion and exclusion criteria

Our analysis focuses on the subset of individuals who become parents for the first time within the observation period. We base this decision on the nature of our empirical strategy: a difference-in-differences around the birth of the first child. As detailed in Section 5, we will exploit the randomness of the sex of the first child to causally estimate the impact of having a daughter, using parents of sons as the control group.

While we could compare parents of daughters to individuals who never become parents, we believe that never-parents may differ from individuals who decide to have children in unobserved ways, making them less comparable. In that case, the parallel trends assumption needed for causal identification may not be plausible.

Focusing on the first childbirth rather than subsequent births ensures a cleaner identification. Later births may be endogenous to earlier experiences of parenthood, including the sex of the first child, and could reflect selective fertility decisions. In contrast, first births offer a more exogenous setting in which to study the causal impact of child sex, minimizing concerns about selection and cumulative family

dynamics.

We restrict to individuals for whom the outcome is available at least once before and once after the first childbirth. Observing the outcome before is necessary to control for pre-treatment level differences across treatment and control groups, whereas observing it after is crucial to estimate the change in outcomes attributable to having a daughter.

To construct the outcome to evaluate H3, we further restrict the sample to respondents for whom partners are identifiable. Given that the outcome is constructed at the couple level and to avoid duplicate observations, we include only female respondents in the regression analysis.

Finally, to ensure robustness of the sample, we restrict our analysis to individuals who were aged 18 to 49 when they first became parents.

5 Empirical strategy

This section describes the empirical strategy used to test the four hypotheses of the study. The first three hypotheses are addressed using a common strategy based on a difference-in-differences design, and centered around the birth of the first child. For H4, which compares the effects across genders, we extend this approach to a triple-differences specification. In all specifications we cluster standard errors at the couple level.

Hypotheses 1–3: Difference-in-differences

To estimate the causal effect of having a daughter (as opposed to a son) on a given outcome, we rely on the fact that the sex of a child is as good as random. The treatment group is defined as as individuals whose first child is a girl, and the control group as those whose first child is a boy. The regression model we plan to estimate is the following:

$$y_{i,t} = \beta_0 + \beta_1 \text{Daughter}_i + \beta_2 \text{Post}_{i,t} + \beta_3 \text{Daughter}_i \times \text{Post}_{i,t} + \boldsymbol{\gamma} \boldsymbol{X}_{i,t} + \epsilon_{i,t} \tag{1}$$

where $y_{i,t}$ is the outcome of interest for individual i at time t, Daughter_i is an indicator equal to one if individual i has a first-born daughter, and $\text{Post}_{i,t}$ is an indicator equal to one for periods after the birth of the first child. The vector $\boldsymbol{X}_{i,t}$ corresponds to the set of controls described in Subsection 3.3.

Hypothesis 4: Triple differences

To test whether the effect of having a daughter differs by parent gender, we estimate a triple-differences model that compares mothers and fathers, allows us to assess whether the treatment effect of having a daughter varies systematically between men and women. The specification of interest is:

$$\begin{aligned} y_{i,t} =& \beta_0 + \beta_1 \text{Man}_i + \beta_2 \text{Daughter}_i + \beta_3 \text{Post}_{i,t} + \beta_4 \text{Man}_i \times \text{Daughter}_i \\ &+ \beta_5 \text{Man}_i \times \text{Post}_{i,t} + \beta_6 \text{Daughter}_i \times \text{Post}_{i,t} + \beta_7 \text{Man}_i \times \text{Daughter}_i \times \text{Post}_{i,t} + \boldsymbol{\gamma} \boldsymbol{X}_{i,t} + \epsilon_{i,t} \end{aligned} \tag{2}$$

where Man_i is an indicator for being male and the remaining variables are as explained above.

6 Interpretation of the results

Before describing the coefficients of interest and explaining the relevant hypothesis tests, we provide an explanation of the size of the coefficients associated with each one of the outcomes. Consider a variable Z and its associated coefficient estimate $\hat{\delta} > 0$. Table 3 provides the magnitude changes in y associated with a unit change in Z according to our regression estimates.

Hypotheses 1–3: Difference-in-differences

The coefficient of interest is β_3, which captures the effect of having a daughter on the outcome. For H1 and H2 the main hypothesis tests whether having a daughter induces a higher outcome compared to having a son corresponding to greater support for gender equality (H1), or more left-leaning political preferences (H2). Formally, these are the null and alternative hypotheses considered, in a one-sided test:

$$\begin{aligned} H_0: &\quad \beta_3 = 0 \\ H_1: &\quad \beta_3 > 0 \end{aligned}$$

A positive and sufficiently large $\hat{\beta}_3$ would suggest that, all else equal, having a daughter leads to a higher level of the outcome among first-time parents compared to having a son. Conversely, a negative coefficient or a coefficient not statistically different from zero would imply that having a daughter is not associated with a higher level of the outcome relative to having a son.

For H3, the main hypothesis should test whether having a daughter induces a smaller deviation from an egalitarian task distribution than having a son. In this case, a higher outcome indicates a larger deviation from equal time use between husband and wife. Thus, the relevant hypothesis to be tested is:

$$\begin{aligned} H_0: &\quad \beta_3 = 0 \\ H_1: &\quad \beta_3 < 0 \end{aligned}$$

A negative and sufficiently large $\hat{\beta}_3$ indicates that, all else equal, having a daughter is associated with a more egalitarian division of tasks in comparison to having a first-born son. A positive or statistically insignificant $\hat{\beta}_3$ would suggest that having a daughter is not associated with higher equality in task division relative to having a son.

Hypothesis 4: Triple differences

We want to assess the following hypothesis:

$$(\text{Father w/ daughter} - \text{Father w/o daughter}) > (\text{Mother w/ daughter} - \text{Mother w/o daughter})$$

To formalize the hypothesis to be tested, it is useful to lay out the predicted post-treatment outcome levels for each group from Equation 2:

- Father with daughter: $\beta_0 + \beta_1 + \beta_2 + \beta_3 + \beta_4 + \beta_5 + \beta_6 + \beta_7$
- Father without daughter: $\beta_0 + \beta_1 + \beta_3 + \beta_5$

- Mother with daughter: $\beta_0 + \beta_2 + \beta_3 + \beta_6$
- Mother without daughter: $\beta_0 + \beta_3$

Formally, the null and alternative hypotheses to be tested are:

$$H_0: \quad \beta_4 + \beta_7 = 0$$
$$H_1: \quad \beta_4 + \beta_7 > 0$$

The coefficient on the child and parent gender interaction, β_4, indicates baseline (pre-birth) differences between fathers of daughters and fathers of sons. Considering that child sex is as good as random, we would expect this coefficient to be statistically zero. The coefficient on the triple interaction term, β_7, captures whether the effect of having a daughter differs for men and women. In other words, it reflects the additional impact of having a daughter after childbirth for fathers, relative to mothers.

A positive and sufficiently large estimate of $\beta_4 + \beta_7$ would indicate that the effect of having a daughter is larger for fathers than for mothers. This would support the hypothesis that men respond more strongly to the sex of their first child than women do. If the sum is negative or not statistically different from zero, we would conclude that there is no larger impact for fathers.

Table 3: Interpretation of the coefficients for different outcome variables.

	Variable	**"All else equal, a unit increase in Z is associated with a $\hat{\delta}$..."**
H1-H4	Best if man/woman work the same to share responsibility	$\times 100$ percentage point higher prob. of agreeing.
H2	Ideological self-placement	standard deviation higher alignment with left-leaning ideology.
H3	Deviation from egalitarian housework	unit higher deviation from egalitarian distribution.

Notes: This table presents the interpretation of the magnitude of the coefficients associated to the outcome variables considered in testing H1-H4. Subsection 3.1 provides an exhaustive description of the construction of all outcomes.

7 Reporting the results

We plan to report one table per hypothesis, each presenting the estimation results obtained using the empirical strategy outlined in Section 5. Each table will contain one row and nine columns. The first and second columns of the tables corresponding to H1 through H3 will report the estimated coefficient $\hat{\beta}_3$ – the interaction effect of having a daughter post-birth – and the corresponding standard error. For H4, the first and second columns of the table will include the estimated $\hat{\beta}_4 + \hat{\beta}_7$ – the differential effect of having a daughter for fathers as specified in Section 5 – and the corresponding standard error.

Each table will also contain one column reporting each of the following statistics: (3) family of the test, (4) residual degrees of freedom, (5) model degrees of freedom, (7) number of observations, and (8) number of participants.

Importantly, each table will contain an additional column (6) displaying the p-value for the hypothesis test corresponding to the main hypothesis of interest, as outlined in Section 6. These p-values will reflect one-sided tests aligned with our directional hypotheses. In H1 and H2, these p-values will correspond to whether the effect of having a daughter is larger than the effect of having a son. In H3,

the reported one-sided test evaluates whether the effect of having a daughter is smaller than the effect of having a son. In the case of H4, the relevant test assesses whether the effect of having a daughter is larger for fathers than for mothers. These additional p-values are included to test our pre-specified theoretical predictions directly and provide a transparent evaluation of the relevant null hypotheses. A final row (9) is included reporting whether the one-sided test is in line with the hypothesis.

References

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